Kongarasan Sathiya Moorthy

AI/ML & GenAI Engineer · M.Sc. Computer Science

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EDUCATION

M.Sc. Computer Science, Universität Passau Current

Focus: AI/ML, Generative AI, RAG, LLM agents

Bachelor's degree, SRM Institute of Science and Technology India

EXPERIENCE

Software Engineer Intern, Triotek Nanocare Solutions Mar 2023 – Feb 2024 · Remote

- Built a single-page web application with HTML, JavaScript and Tailwind CSS from user research and high-fidelity UI designs.
- Deployed backend packages and configured an Nginx reverse proxy to serve the production API.
- Contributed to the UI redesign and React development of the company's mobile app.

PROJECTS

RAG Assistant: Document Q&A 2026

- Full-stack RAG app with a FastAPI backend and retrieval pipeline (chunking, embeddings, FAISS), LLaMA 3.3 70B via Groq and LangChain, and a React frontend; tuned chunking and retrieval for grounding.

AirNetBus: 8th TUM Science Hackathon (Airbus challenge) Jun 2026

- Built a decentralised AI coordination platform for autonomous eVTOLs in 48 hours, with a real-time digital twin of Munich and multi-objective routing around weather, noise and traffic.

Audio-Reactive Resonance Landscape (Universität Passau) 2026

- Python pipeline converting orchestral music into cinematic video: librosa/NumPy/SciPy feature extraction feeding a GLSL shader rendered with ModernGL and exported with FFmpeg.

AI Resume Analyzer · AI Workflow Automation 2025

- An LLM evaluation app (FastAPI, Streamlit, Gemini) with NLP pipelines and synthetic datasets for reproducible robustness experiments.
- Multi-step LLM automation pipelines with n8n and REST APIs, with benchmarking to select model configurations.

Book Data Integration 2025

- Schema and entity matching across Amazon, Goodreads and Kindle data with six similarity matchers and combiners, evaluated with precision, recall and F1.

OpenEYE · Woodie (SRMIST) 2022 – 2024

- OpenEYE: assistive app with YOLOv8 object recognition and InceptionV3 + Transformer captioning, built in React Native.
- Woodie: campus sustainability platform (React, Three.js, Node.js, Kotlin, Nginx) deployed and used at SRM University.

SKILLS

AI/ML: PyTorch, LangChain, RAG, LLM agents, FAISS, YOLOv8, Transformers, spaCy, NLP

Backend & Data: Python, FastAPI, SQL, Node.js, REST APIs, Pandas, NumPy, SciPy

Frontend & Mobile: React, React Native, Three.js, Tailwind CSS, Kotlin, Figma

Tools: Git, Nginx, n8n, FFmpeg, GLSL/ModernGL, Jupyter

CERTIFICATIONS

8th TUM Science Hackathon, TUM: Junge Akademie (2026) · Introduction to Mobile Development, Meta (2023) · AI for Everyone, Coursera (2022) · Cloud Computing, UpSkillz Solutions